

**Exercise 1 - Trace as the sum of eigenvalues.** Given any square matrix  $M$ , with eigenvalues  $\{\lambda\}_{i=1}^n$ , then

$$\text{Tr}(M) \stackrel{\text{def}}{=} \sum_{i=1}^n M_{ii} = \sum_{i=1}^n \lambda_i. \quad (1)$$

In this exercise we are going to show this property for symmetric matrices.

1. Prove the *cyclic property* of the trace: given  $A \in \mathbb{R}^{m \times n}$ ,  $B \in \mathbb{R}^{n \times m}$ , then  $\text{Tr}(AB) = \text{Tr}(BA)$ <sup>1</sup>
2. Leveraging the cyclic property, show (1). (*hint: recall the Spectral Theorem.*)

**Exercise 2 - Frobenius norm.** Given any matrix  $A \in \mathbb{R}^{m \times n}$  with elements  $A_{ij}$ , the Frobenius norm of  $A$  is defined as:

$$\|A\|_F^2 \stackrel{\text{def}}{=} \sum_{i=1}^m \sum_{j=1}^n A_{ij}^2. \quad (2)$$

1. Show that the Frobenius norm can be also defined as  $\sqrt{\text{Tr}(A^T A)}$ .
2. Prove that, if  $\{\sigma_i\}_{i=1}^{\min\{m,n\}}$  are the singular values of  $A$ , then

$$\|A\|_F^2 = \sum_{i=1}^{\min\{m,n\}} \sigma_i^2. \quad (3)$$

**Exercise 3 - Orthogonal matrices are rotations/reflections.** Rotation and reflections are linear transformations preserving the norm of vectors and the angles between them.

Consider an orthogonal matrix  $U \in \mathbb{R}^{n \times n}$  and prove the following statements:

1.  $\|Uv\|_2 = \|v\|_2$ , for any vector  $v \in \mathbb{R}^n$ . (norm preservation)
2.  $\langle Uv, Uw \rangle = \langle v, w \rangle$ , for any  $v, w \in \mathbb{R}^n$ . (angle preservation)
3. If  $\sigma$  is a singular value of  $U$ , then  $\sigma = 1$ .
4. If  $\lambda$  is an eigenvalue of  $U$ , then  $|\lambda| = 1$ . (*hint: leverage the norm preservation property*)

**Exercise 4 - Projections.** Consider the projection matrix  $P \in \mathbb{R}^{n \times n}$ . Show that following statements are true:

1.  $P_\perp \stackrel{\text{def}}{=} I_n - P$  defines a projection.
2. Defining  $v_\perp \stackrel{\text{def}}{=} P_\perp v$  for any vector  $v \in \mathbb{R}^n$ , then  $Pv_\perp = 0$ .
3. Any vector  $v \in \mathbb{R}^n$  can be written as  $v = u + Pv$ , where  $u \in \ker(P)$ .
4. If  $\lambda$  is an eigenvalue of  $P$ , then  $\lambda = 0$  or  $\lambda = 1$ .
5. If the projection  $P$  is also symmetric, then  $\text{Tr}(P) = \text{rank}(P)$ .

**Exercise 5 - Pseudoinverse.** Consider a matrix  $A \in \mathbb{R}^{m \times n}$  whose SVD is defined as

$$A = \sum_{i=1}^r \sigma_i u_i v_i^T$$

where  $r \leq \min(m, n)$  is the rank of  $A$  and  $\sigma_i > 0$  are the positive singular values, with  $(u_i)_{i=1}^r$  and  $(v_i)_{i=1}^r$  being two families of orthogonal vectors of  $\mathbb{R}^m$  and  $\mathbb{R}^n$ , respectively.

1. Show that there are two orthogonal matrices  $U \in \mathbb{R}^{m \times m}$  and  $V \in \mathbb{R}^{n \times n}$  such that

$$A = U \begin{pmatrix} \Sigma & 0_{r \times (n-r)} \\ 0_{(m-r) \times r} & 0_{(m-r) \times (n-r)} \end{pmatrix} V^T,$$

where  $\Sigma$  is the  $r \times r$  diagonal matrix whose entries are the  $\sigma_i$ .

2. Show that the pseudoinverse of  $A$  can be written as

$$A^+ = V \begin{pmatrix} \Sigma^{-1} & 0_{r \times (m-r)} \\ 0_{(n-r) \times r} & 0_{(n-r) \times (m-r)} \end{pmatrix} U^T,$$

3. Assume that  $A$  is full rank, ie  $r = \min(m, n)$ . Verify that the pseudoinverse can be written as

$$A^+ = \begin{cases} (A^T A)^{-1} A^T & \text{if } m \geq n, \\ A^T (A A^T)^{-1} & \text{if } m < n. \end{cases}$$

4. Show that  $AA^+$  and  $A^+A$  are orthogonal projectors.

<sup>1</sup>This property can be easily extended to the product of any arbitrary number of matrices and all their cyclic permutations, e.g.

$\text{Tr}(ABCD) = \text{Tr}(DABC) = \text{Tr}(CDAB) = \text{Tr}(BCDA)$ .